Build a Personalized Online Course Recommender System with Machine Learning

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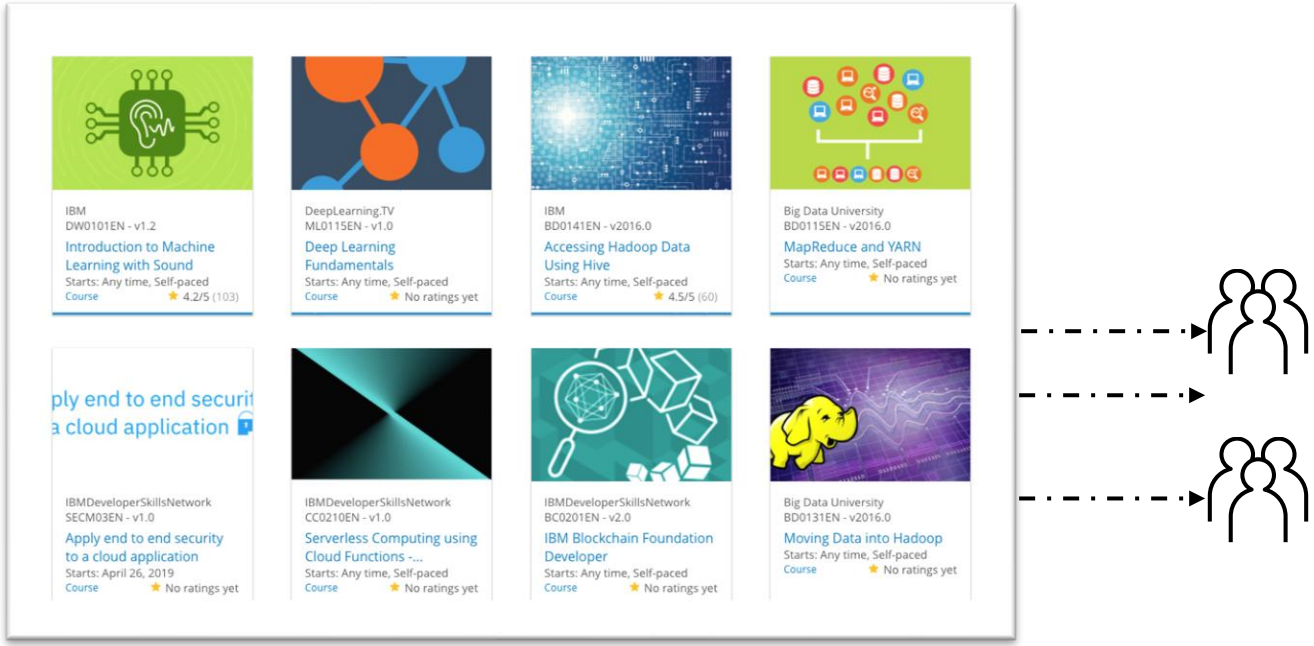


Outline

- Introduction and Background
- Exploratory Data Analysis
- Content-based Recommender System using Unsupervised Learning
- Collaborative-filtering based Recommender System using Supervised learning
- Conclusion
- Appendix

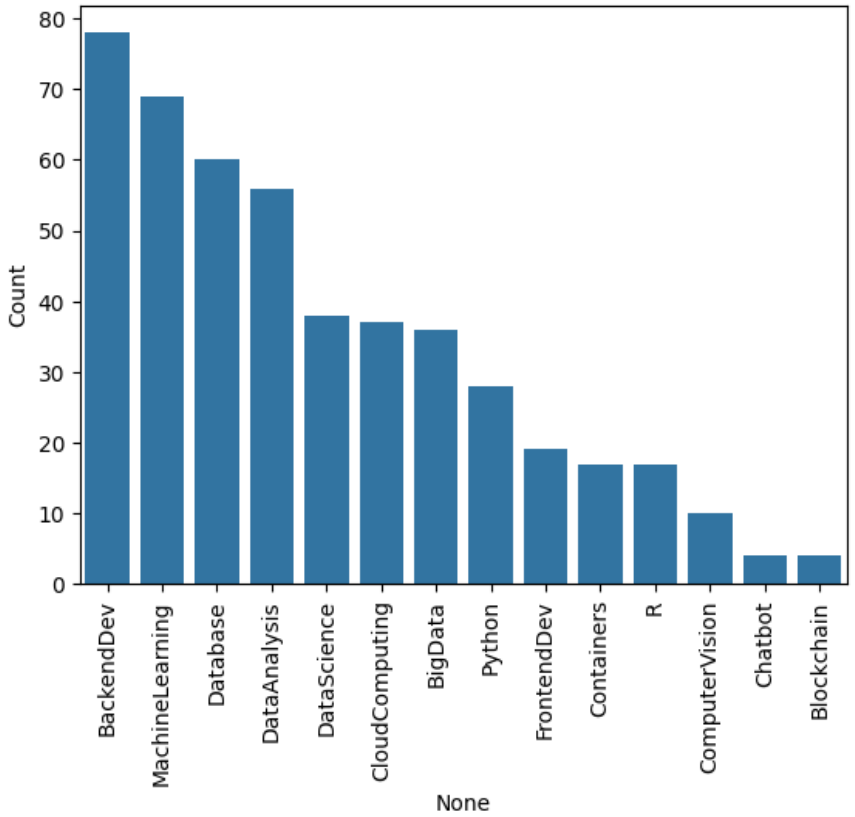
Introduction

- Learner across the world increasingly seek to acquire skills in leading technologies such as Machine Learning, Ai, Data Science, Cloud, App Development, etc, However with the growing number of available courses, it becomes difficult for the learners to identify relevant content efficiently. To improve the learning experience a recommender system can users discover aligned with their interests and effectively guide their learning path.
- Currently, learners struggle to discover relevant courses due to the large volume of available content. This led to inefficient course selection and reduced engagement, ultimately affecting learning outcomes and platform effectiveness.

Exploratory Data Analysis

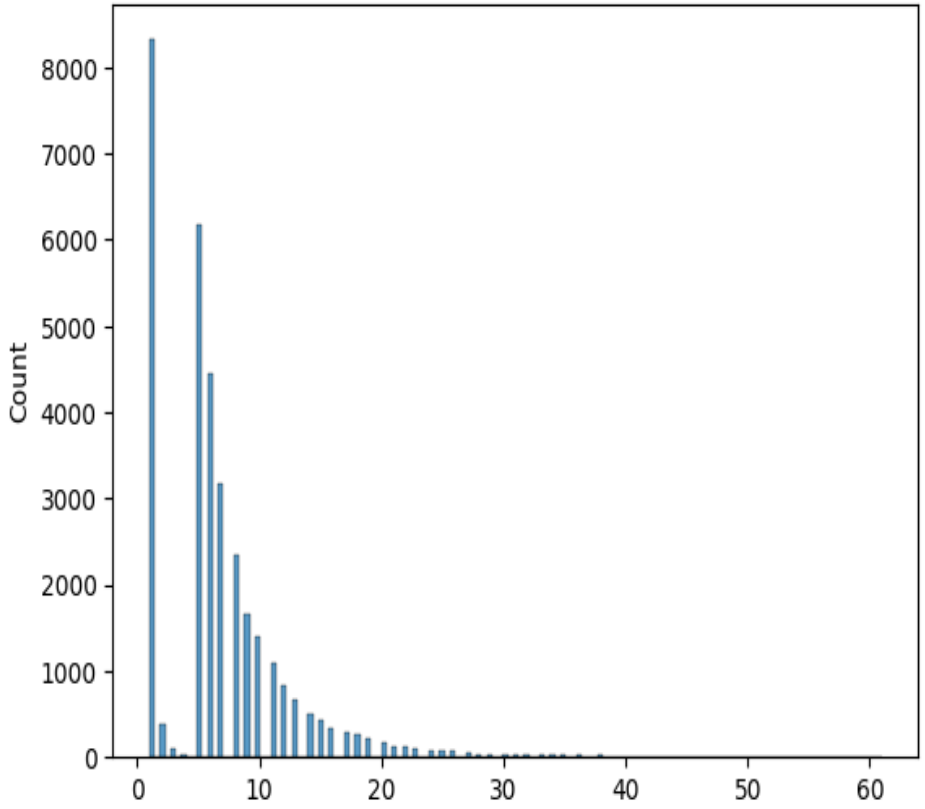
Course counts per genre

- This chart shows the distribution of courses enrollments across different categories.
- The x axis represent the courses genre, and the y axis show the number of enrollments.
- We can see that BackendDev has the highest enrollments and the Blcokchain has the lowest.
- Following the trend we can see the learners are more interstead in Data, ML and Backend skills.



Course enrollment distribution

- This chart shows the distribution of courses enrollments across users.
- The x axis represent the courses number, and the y axis show the number of users (count).
- most users are concentrated below 10 courses.
- The distribution is skewed to the right indicating that while most users take few courses, a small number of users enroll in many courses



20 most popular courses

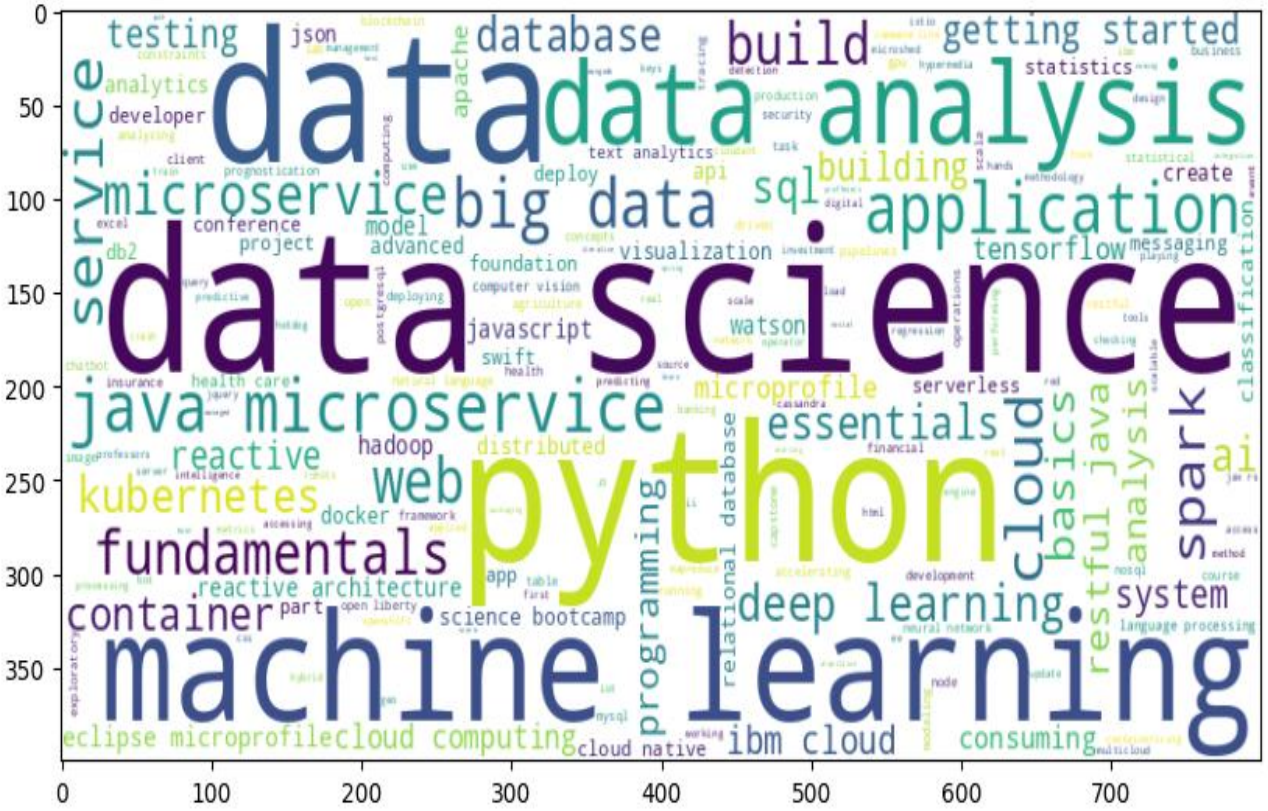
- The top 20 courses show a clear concentration of enrollments in a few highly popular courses, with the top courses exceeding 13000 enrollments.
- After the top 5, there is noticeable drop, with enrollments decreasing to around 8000 and below.
- This indicate that a small number of courses dominate users interest.

20 rows 20 rows × 2 cols

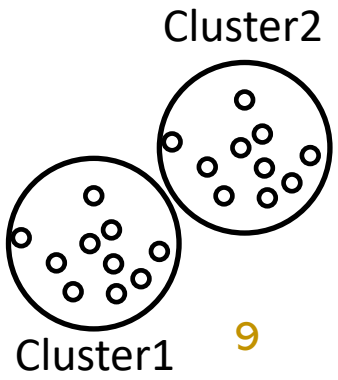
	course	Ratings
0	PY0101EN	14936
1	DS0101EN	14477
2	BD0101EN	13291
3	BD0111EN	10599
4	DA0101EN	8303
5	DS0103EN	7719
6	ML0101ENv3	7644
7	BD0211EN	7551
8	DS0105EN	7199
9	BC0101EN	6719
10	DV0101EN	6709
11	ML0115EN	6323
12	CB0103EN	5512
13	RP0101EN	5237
14	ST0101EN	5015
15	CC0101EN	4983
16	CO0101EN	4480
17	DB0101EN	3697
18	BD0115EN	3670
19	DS0301EN	3624

Word cloud of course titles

- This word cloud represent the most frequent keywords across the courses titles and descriptions, larger words indicate higher frequency and importance.
- We can observe that words like data science, data, python and machine learning dominate, suggesting a strong focus on data-driven and AI related topics.

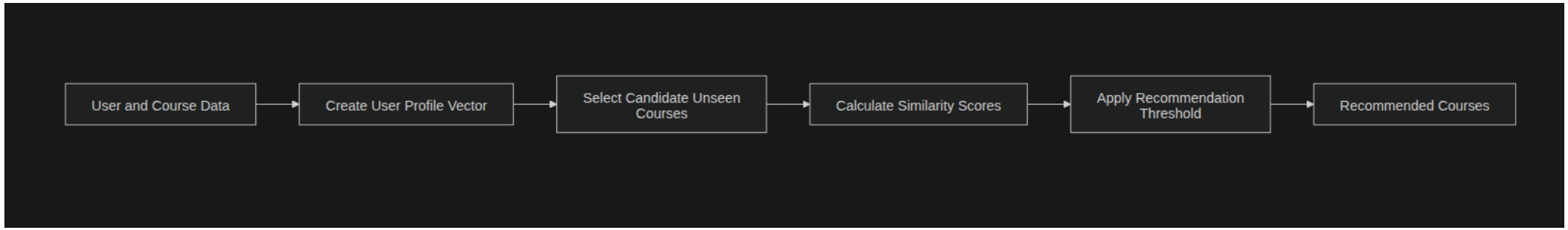


Content-based Recommender System using Unsupervised Learning



Flowchart of content-based recommender system using user profile and course genres

- Starting by getting the user data then creating vectors for each profile after that selecting the unseen candidates, then we calculate the similarity scores, applying the recommendation threshold, finally getting the recommended scores.



Evaluation results of user profile-based recommender system

- Recommendation Score Threshold: 10.0

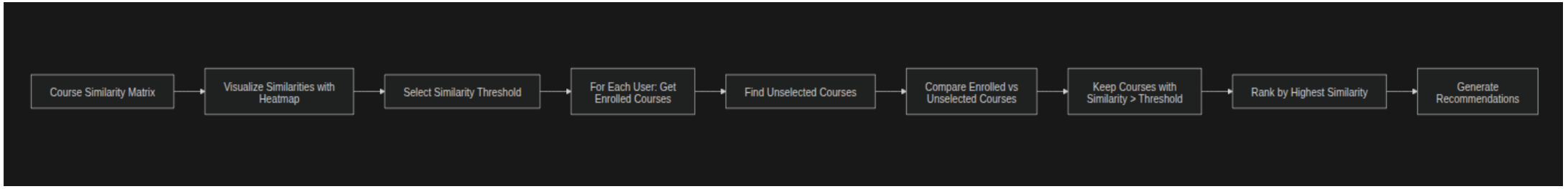
- On average, 60 courses have been recommended per user.

- the top-10 commonly recommended courses across all users

TA0106EN	17378
excourse22	15639
excourse21	15639
GPXX0IBEN	15632
ML0122EN	15597
excourse06	15048
excourse04	15048
GPXX0TY1EN	14677
excourse31	14445
excourse72	14439

Flowchart of content-based recommender system using course similarity

- Starting by getting the course similarity Matrix then visualizing heatmap, select threshold, after that for each user selecting the unseen candidates, then we calculate the similarity scores, applying the recommendation threshold, finally getting the recommended scores.



Evaluation results of course similarity based recommender system

- Recommendation Score Threshold: 0.6

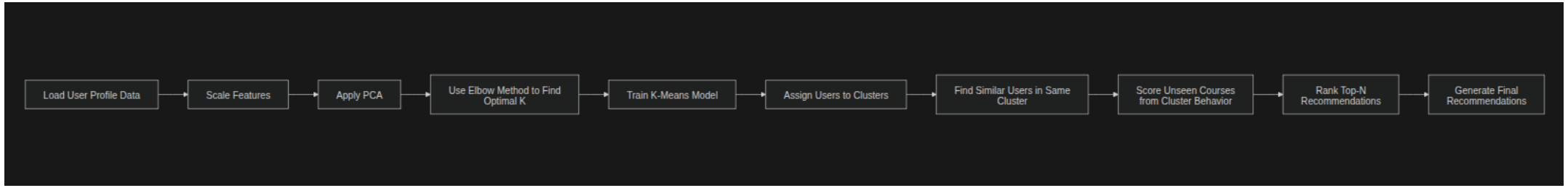
- On average, 9.7 courses have been recommended per user.

- the top-10 commonly recommended courses across all users

DS0110EN	15003
excourse22	14937
excourse62	14937
excourse63	14641
excourse65	14641
excourse68	13551
excourse72	13512
excourse74	13291
excourse67	13291
BD0145EN	12497

Flowchart of clustering-based recommender system

- The flow starts by preprocessing user data through scaling and dimensionality reduction (PCA), followed by selecting the optimal number of clusters using the elbow method and training a K-Means model. Users are then grouped into clusters, where similar users are identified based on cluster membership. Finally, recommendations are generated by ranking unseen courses according to their popularity within each user's cluster.



Evaluation results of clustering-based recommender system

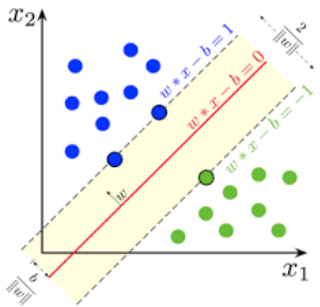
- PCA variation: 0.95, KMeans n_init: 10, n_clusters: 9

- On average, each user receives approximately 4.89 new (unseen) course recommendations out of the top 5 recommendations.

- the top-10 commonly recommended courses across all users

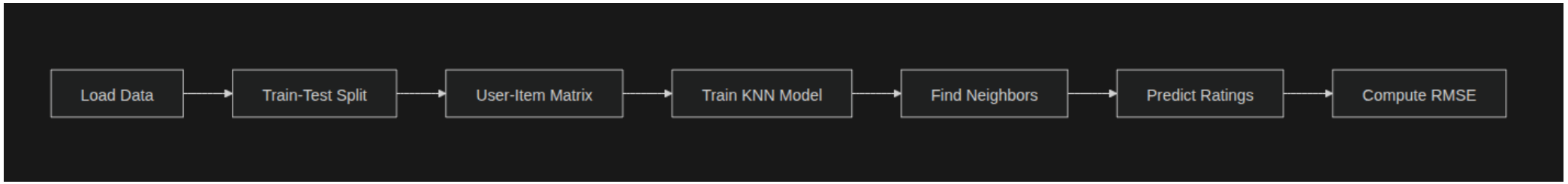
course	count
Blockchain	20744
CloudComputing	18649
MachineLearning	16820
BigData	16602
Database	14318
BackendDev	11753
R	11632
DataScience	10252
Containers	9356
DataAnalysis	8872

Collaborative-filtering Recommender System using Supervised Learning



Flowchart of KNN based recommender system

- This chart start by transforms rating data into a user–item matrix, then uses cosine similarity to find similar users and predict ratings based on their preferences. The model's performance is evaluated using RMSE by comparing predicted ratings with actual values.

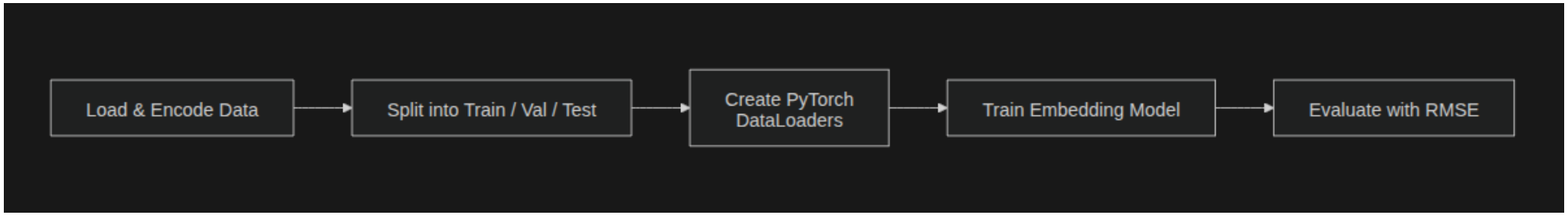


Flowchart of NMF based recommender system

- This flowchart shows a matrix factorization recommender system that learns user and item latent features through training. It predicts ratings using the dot product of learned vectors and evaluates performance using RMSE.

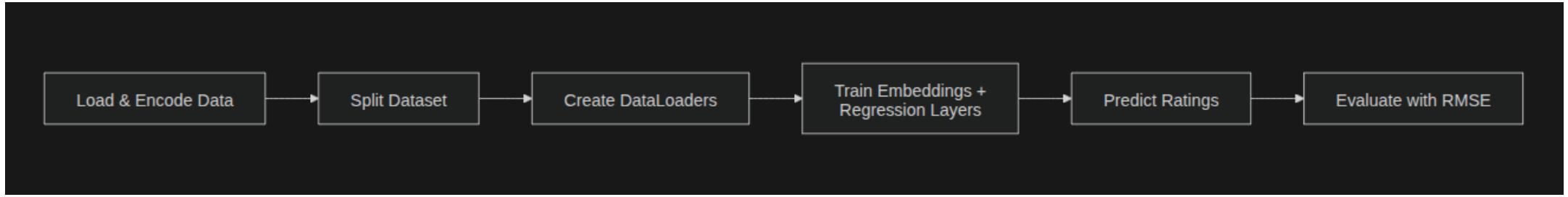
Flowchart of Neural Network Embedding based recommender system

- This flowchart shows an embedding-based recommender system built with PyTorch. It encodes users and items, learns their latent representations through training, and predicts ratings using embeddings and bias terms. The model is evaluated on unseen data using RMSE.



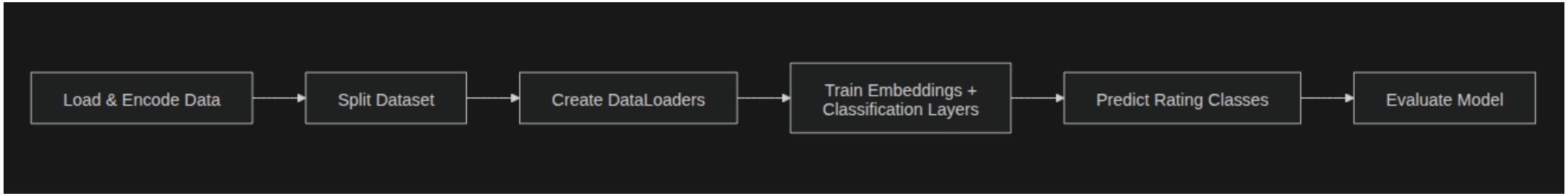
Flowchart of Neural Network Embedding-Regression based recommender system

- This flowchart represents a neural recommender system that combines user and item embeddings with additional regression layers to model complex interactions. The concatenated embeddings are passed through a neural network to predict ratings. The model is evaluated using RMSE on the test set.



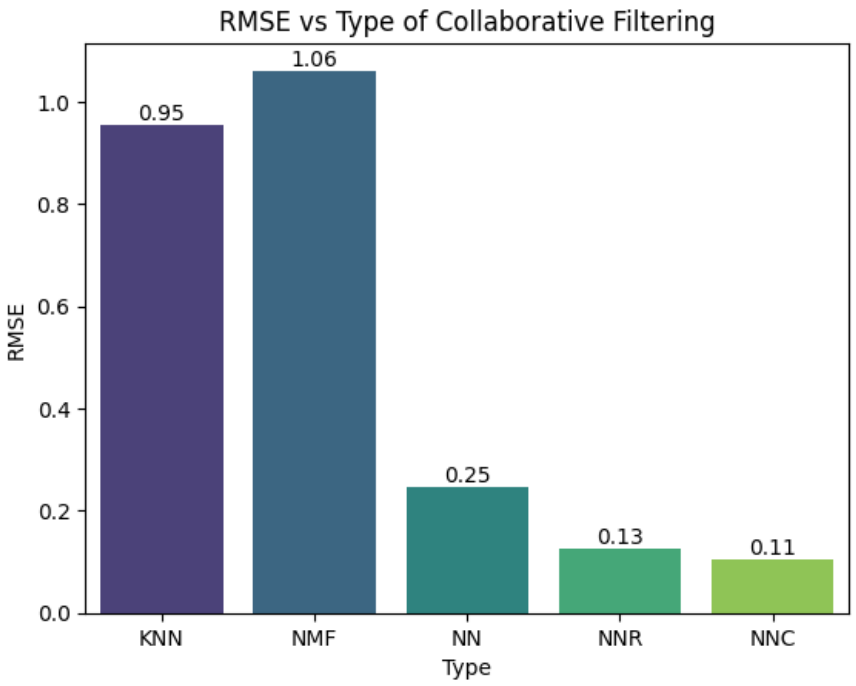
Flowchart of Neural Network Embedding-Classification based recommender system

- This flowchart represents a neural recommender system that uses user and item embeddings combined with classification layers to predict rating categories. The model learns to classify interactions instead of regressing exact values. Performance is evaluated based on prediction accuracy or related classification metrics.



Compare the performance of collaborative-filtering models

- This chart compares the RMSE performance of different collaborative filtering approaches. Traditional methods like KNN and NMF show higher error, while neural models significantly improve accuracy. The best performance is achieved by neural architectures, especially the classification-based model (NNC), indicating their superior ability to capture complex user–item interactions.



Conclusions

- Users struggle to discover relevant courses due to content overload.
- Content based, collaborative filtering, and deep learning models are evaluated approaches.
- User preferences are highly nonlinear and context dependent.
- Traditional Models like KNN provide baseline performance, but neural models significantly improve prediction accuracy.
- Embedding-based with Classification neural architecture achieved the best results.
- Cold start problem is a limitation for new Users/Items.
- For future work a hybrid recommender system can give better results.

Appendix

- Github Repository for the Code: Link